

Technology Stack

Date	28 September 2026
Team ID	6ab9f220c23107aaaa73f984
Project Name	ComicCraft AI - AI comic story creator

Technology Stack

Layer	Technology	Purpose
Story generation	Google Gemini Flash (google-genai SDK)	Title, panels, dialogue, captions, image prompts as structured JSON
Image generation	Stable Diffusion XL 1.0 via Stability AI API	Comic panel artwork (stable-diffusion-xl-1024-v1-0)
Backend	Python 3.10+, FastAPI, Uvicorn	REST endpoints and page serving
Validation	Pydantic	Request and comic data models
PDF export	ReportLab	Storyboard PDF
Image handling	Pillow	Image validation and placeholder panels
HTTP client	requests	Calls to the Stability AI API
Frontend	HTML5, CSS, vanilla JavaScript	Form, progress, comic layout, redraw
Templating	Jinja2	index.html, comic_preview.html, export_success.html
Config	python-dotenv	Loads keys and settings from .env

Architecture note

The app uses a two-stage AI pipeline: Gemini writes the script, then Stable Diffusion XL draws each panel. Text is kept out of the images on purpose; the web page draws dialogue and captions so they stay readable.